



LINMA2415 Quantitative Energy Economics

ISSAMBRE L'HERMITE DUMONT

This summary may not be up-to-date, the newer version is available at this address:
<https://github.com/SimonDesmidt/Syntheses>

Academic year 2025-2026 - Q2



UCLouvain

Contents

1	Reminder about optimization	2
1.1	Formulations	2
1.2	Optimality conditions	3
2	Power systems	4
2.1	Production	4
2.2	Transmission and distribution	6
2.3	Consumption	6
2.4	Reserves	7
2.5	Multi-stage operations	7
3	Economic Dispatch	9
3.1	Formulation	9
3.2	Competitive market equilibrium	10
4	Power markets	12
4.1	Exchange VS Pool market	12
4.2	Electricity auctions	13
4.3	Nodal VS Zonal pricing	13
5	Power flow	15
5.1	Network modeling	15
5.2	Circuits	15
5.3	The power flow equations	16
5.4	DC power flow using bus angles	16
5.5	DC power flow using lines flows	16
5.6	DC optimal power flow	16
6	Pricing transmission	19
6.1	Nodal pricing	20
6.2	Zonal pricing	20
6.3	Other formulations	21
7	Unit commitment and nonconvexities	23
7.1	The unit commitment problem	23
7.2	Market equilibrium under non-convexities	24

Reminder about optimization

1.1 Formulations

Definition 1.1. The primal problem is the original optimization problem that we want to solve. It is formulated as:

$$\begin{aligned} \max_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned} \tag{1.1}$$

Definition 1.2. The dual problem is derived from the primal problem. It is formulated as:

$$\begin{aligned} \min_y \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned} \tag{1.2}$$

Another way to look at the dual problem is by using the Lagrangian function. Consider now the primal under the form:

$$\begin{aligned} p^* = \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p \end{aligned} \tag{1.3}$$

Definition 1.3. The Lagrangian function is defined as:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x) \tag{1.4}$$

Definition 1.4. The Lagrange dual function is defined as:

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) \tag{1.5}$$

Definition 1.5. The Lagrange dual problem is defined as:

$$\begin{aligned} d^* = \max \quad & g(\lambda, \nu) \\ \text{s.t.} \quad & \lambda \geq 0 \end{aligned} \tag{1.6}$$

1.2 Optimality conditions

Definition 1.6. Weak duality is when $d^* \leq p^*$, it always holds and can be used to find bounds on the optimal value of the primal problem.

Definition 1.7. Strong duality is when $d^* = p^*$, does not holds in general, but usually holds for convex problems.

Definition 1.8. Complementary slackness is a condition that holds at optimality, it states that for each constraint, either the constraint is active (i.e. $f_i(x) = 0$) or the corresponding dual variable is zero (i.e. $\lambda_i = 0$).

$$0 \leq \lambda_i^* \perp f_i(x^*) \leq 0 \quad (1.7)$$

Definition 1.9. The KKT conditions are a set of necessary conditions for optimality. Consider this problem:

$$\begin{aligned} \max \quad & f(x, y) \\ \text{s.t.} \quad & Ax + By \leq b \quad (\lambda) \\ & Cx + Dy = d \quad (\mu) \\ & x \geq 0 \quad (\lambda_2) \end{aligned} \quad (1.8)$$

The KKT conditions are:

$$\begin{aligned} & Cx + Dy - d = 0 \\ & 0 \leq \lambda \perp Ax + By - b \leq 0 \\ & 0 \leq x \perp \lambda^T A + \mu^T C - \nabla_x f(x, y)^T \geq 0 \\ & \lambda^T B + \mu^T D - \nabla_y f(x, y)^T = 0 \end{aligned} \quad (1.9)$$

And the Lagrangian function can be written as:

$$L(x, y, \lambda, \mu) = f(x, y) + \lambda^T (b - Ax - By) + \mu^T (d - Cx - Dy) + \lambda_2^T x \quad (1.10)$$

Power systems

In our world, everything works with electricity and to be able to use it, there is complex systems behind it. In this chapter, we will see how the electricity is produced, transmitted and consumed. Usually a power system work like this: Electricity is produced in power plants, then it is transmitted through high voltage lines to substations, where it is transformed to lower voltage and distributed to consumers. The electricity is consumed by households, industries and other consumers. To ensure that the electricity supply meets the demand, there are reserves that can be activated in case of a sudden increase in demand or a decrease in supply. All of this electricity is exchanged through financial markets, where producers and consumers can buy and sell electricity and other services.

2.1 Production

Electricity can be produced in different ways, there is two main categories of production: renewable and non-renewable. Renewable energy sources are those where the "fuel" is naturally present and is conceptually infinite, such as solar, wind, hydro, geothermal and biomass. Non-renewable energy sources are those where the "fuel" is extracted from the earth and is finite, such as coal, oil and natural gas. Nuclear energy is also considered non-renewable, although it does not emit greenhouse gases. Each type of production has its own characteristics and costs, and they are all important for the functioning of the power system. We can measure electricity by two main quantities: power and energy.

Definition 2.1. Power (MW) is the rate at which electricity is produced or consumed at a given moment. It is the instantaneous amount of electricity being generated or used.

Definition 2.2. Energy (MWh) is the total amount of electricity produced or consumed over a period of time. It is the integral of power over time.

Every type of production has its own structure and thus obey to different constraints to produce electricity. For example, here are some of the constraints that we can find in different types of production:

- **Fossil fuel power plants:**
 - Min/max power output
 - Ramp constraints (how fast the power can be increased or decreased)
 - Min up/down time (how long the plant must be on or off before it can change state)

- **Storage:**
 - Max charge/discharge power
 - Energy capacity (how much energy can be stored)
- **Renewables:**
 - Nameplate capacity (max power output under ideal conditions)
 - Capacity factor (actual power output as a fraction of nameplate capacity, depends on weather conditions)

But the production of electricity is not free. There are different types of costs associated with the production of electricity, such as fuel costs, fixed costs etc.

Definition 2.3. Variable cost (€/h) is the cost that depends on the amount of electricity produced. For example, fuel costs.

Definition 2.4. Fixed cost (€) is the cost that does not depend on the amount of electricity produced. For example, construction costs (€), overnight cost (€/kW (invested kW)).

Definition 2.5. Marginal cost (€/MWh) is the cost of producing one additional megawatt-hour of electricity. It is the derivative of the variable cost with respect to the power output.

Definition 2.6. Average cost (€/MWh) is the total cost per unit of electricity outputed.

Definition 2.7. The merit order curve is a graph that shows the marginal cost of production for different types of power plants. It is used to determine which power plants should be used to meet the demand at a given time, with the cheapest ones being used first.

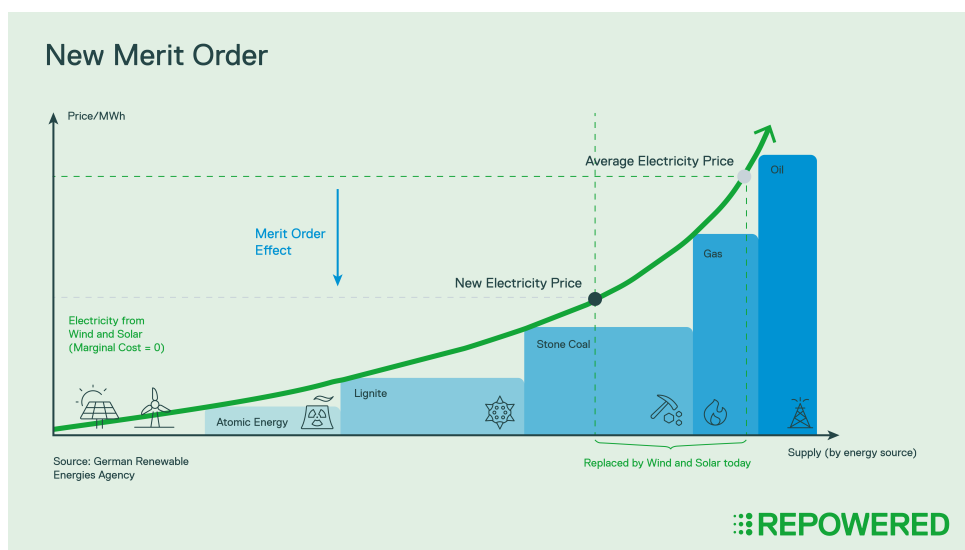


Figure 2.1: Example of merit order curve

2.2 Transmission and distribution

Definition 2.8. **Transmission** is the process of transporting electricity from the power plants to the substations. It is done through high voltage lines.

Definition 2.9. **Distribution** is the process of transporting electricity from the substations to the consumers. It is done through low voltage lines.

The transmission and distribution of electricity can be represented by a graph.

Definition 2.10. **Buses** are the nodes of the graphs, they represent the substations, producers and consumers. The power should be balanced at each bus, meaning that the power injected by the producers should be equal to the power consumed by the consumers and the power flowing through the lines connected to the bus.

Definition 2.11. **Lines** are the edges of the graph, they represent the transmission lines. They have a can have capacity.

The electricity flows through the lines according to the laws of physics, and the flow on each line is determined by the power injections at the buses and the impedance of the lines.

Definition 2.12. **Power flow equations** determine a mapping $f = P(r)$ of power injections r at the buses to power flows f on the lines.

Definition 2.13. **DC power flow equations** are an approximation of the power flow equations $f = P(r)$ by a linear mapping.

2.3 Consumption

Like the production of electricity, the consumption of electricity also has its own characteristics.

Definition 2.14. **Consumer benefit** (€/h) is the benefit that a consumer gets from consuming a certain amount of electricity.

Definition 2.15. **Marginal benefit** (€/MWh) is the benefit of consuming one additional megawatt-hour of electricity. It is the derivative of the consumer benefit with respect to the power consumption.

Definition 2.16. **Inverse demand / marginal benefit function** maps power consumption Q to marginal benefit $MB(Q)$.

Definition 2.17. **Demand function** maps prices of power v to quantity Q , $Q(v)$. It is the inverse mapping of the inverse demand function.

Definition 2.18. **Elasticity** is the sensitivity of the demand $Q(v)$ to changes in price v . It is defined as:

$$\epsilon = \frac{\frac{dQ}{dv}}{\frac{Q}{v}} \quad (2.1)$$

Definition 2.19. **Average Value of Lost Load** is the average cost to consumers of not having electricity. It is used to estimate the cost of power outages and to determine the value of reliability in the power system.

Definition 2.20. **Value of Lost Load (VOLL)** is the cost to consumers of not having electricity for a specific amount of time.

2.4 Reserves

With the rise of renewable energy sources, the uncertainty in the power system has increased, and the need for reserves has become more important. **Contingencies** (failures of components in the power system) can cause sudden changes in the supply or demand of electricity, and reserves are needed to ensure that the power system can still operate reliably. There are different types of reserves, that can be activated in different time frames:

- **Primary reserves** are activated and operational within seconds to few minutes after a contingency, they are used to stabilize the frequency of the power system. It is the first line of defense. It is usually provided by batteries for example because they are not limited by ramp constraints and can respond very quickly.
- **Secondary reserves** are activated within seconds but are only fully available within 5-10 minutes after a contingency. They can be supplied by spinning (on-line generators) or non-spinning reserves (offline generators but can be brought online quickly).
- **Tertiary reserves** are available within 15 minutes.

These reserves are activated sequentially. And on the following graph, we can see how the different reserves have been activated over a day in the Belgian power system.

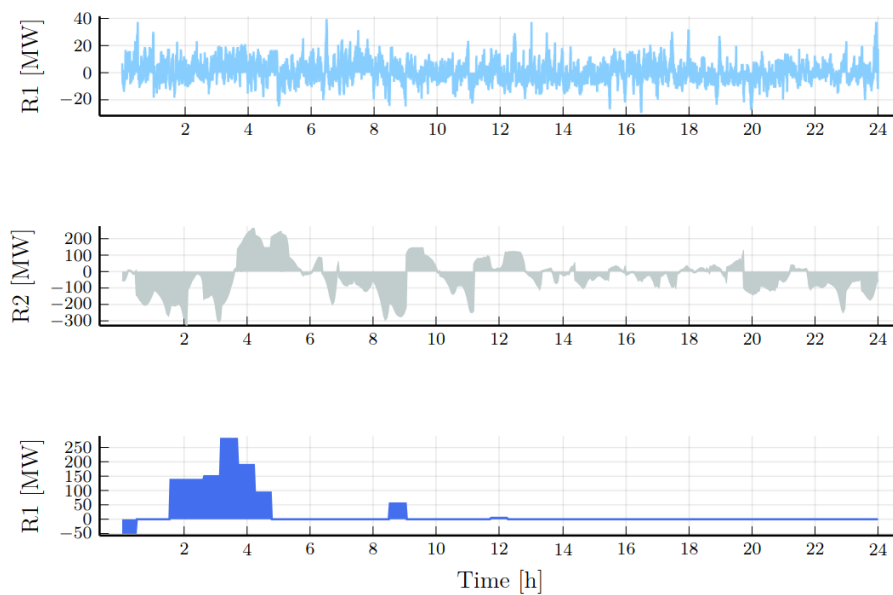


Figure 2.2: Example of activation of reserves in the Belgian power system

2.5 Multi-stage operations

The financial markets for electricity are organized in different time frames with different products exchanged at each time frame. This can be resumed in the following figure:

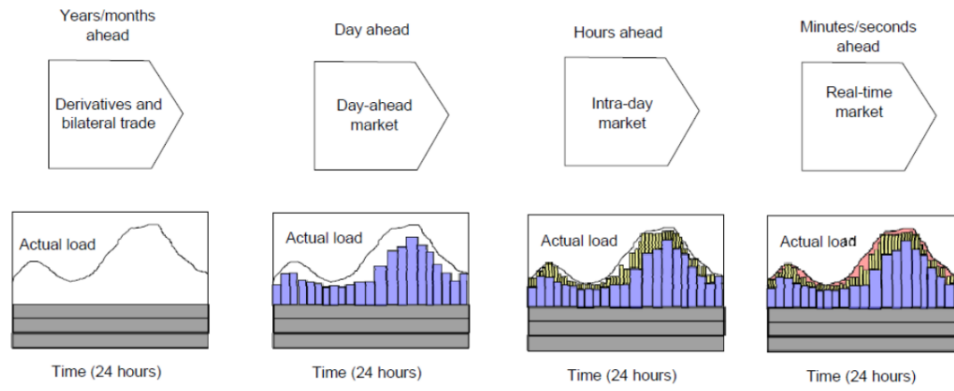


Figure 2.3: Order of the markets for electricity

Economic Dispatch

The economic dispatch problem is the problem of determining the optimal production of electricity by the different power plants to meet the demand at the lowest cost, while respecting the constraints of the power system. It is the simplest model that models a power system.

3.1 Formulation

The economic dispatch problem aims to maximize the total welfare of the system, which is the total benefit of the consumers minus the total cost of the producers. It can be formulated as follows:

$$\begin{aligned}
 \max \quad & \sum_{l \in L} \int_0^{d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\
 \text{s.t.} \quad & \sum_{l \in L} d_l - \sum_{g \in G} p_g = 0 \quad [\lambda] \\
 & 0 \leq d_l \leq D_l, \quad \forall l \in L \quad [\nu_l] \\
 & 0 \leq p_g \leq P_g, \quad \forall g \in G \quad [\mu_g]
 \end{aligned} \tag{3.1}$$

Where L is the set of loads, G is the set of generators. MC_g is the increasing marginal cost and MB_l is the decreasing marginal benefit. The KKT conditions for this problem are:

$$\begin{aligned}
 0 &\leq p_g \perp MC_g(p_g) - \lambda + \mu_g \geq 0 \\
 0 &\leq \mu_g \perp P_g - p_g \geq 0 \\
 0 &\leq d_l \perp -MB_l(d_l) + \lambda + \nu_l \geq 0 \\
 0 &\leq \nu_l \perp D_l - d_l \geq 0 \\
 \lambda &\perp \sum_{g \in G} p_g - \sum_{l \in L} d_l = 0
 \end{aligned} \tag{3.2}$$

There exists a unique price λ (except in some cases) that clears the market, meaning that the supply equals the demand. This price is called the **market clearing price** and it is the price at which the total welfare is maximized. We can find this price by solving the economic dispatch problem or by using the KKT conditions. So there exists a threshold λ such that:

- If $0 < p_g < P_g$, then $MC_g(p_g) = \lambda$.
- If $0 < d_l < D_l$, then $MB_l(d_l) = \lambda$.

- If $p_g = 0$, then $MC_g(p_g) \geq \lambda$.
- If $d_l = 0$, then $MB_l(d_l) \leq \lambda$.
- If $p_g = P_g$, then $MC_g(p_g) \leq \lambda$.
- If $d_l = D_l$, then $MB_l(d_l) \geq \lambda$.

3.2 Competitive market equilibrium

Definition 3.1. A market is **perfectly competitive** if

- Agent are price takers, meaning that they cannot influence the market clearing price λ .
- Variable costs are convex and benefits are concave.
- Agent have access to public information.

Definition 3.2. The **aggregate cost** is the cheapest way to produce Q MW of power among a set of producers. It is defined as:

$$\begin{aligned} TC_G(Q) = \min \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\ \text{s.t.} \quad \sum_{g \in G} p_g = Q \\ p_g \in \text{dom}(MC_g) \end{aligned} \quad (3.3)$$

And thus the **aggregate marginal cost** is $MC_G(Q) = TC'_G(Q)$

Definition 3.3. The **aggregate benefit** is the cheapest way to consume Q MW of power among a set of consumers. It is defined as:

$$\begin{aligned} TB_L(Q) = \max \sum_{l \in L} \int_0^{d_l} MB_l(x) dx \\ \text{s.t.} \quad \sum_{l \in L} d_l = Q \\ d_l \in \text{dom}(MB_l) \end{aligned} \quad (3.4)$$

And thus the **aggregate marginal benefit** is $MB_L(Q) = TB'_L(Q)$. In a competitive market, the market clearing price λ is decided with the **market clearing condition**:

$$\lambda \perp \sum_{g \in G} p_g - \sum_{l \in L} d_l = 0 \quad (3.5)$$

Definition 3.4. A market is at **equilibrium** if

- No profitable opportunities for trade exist, meaning that no agent can increase their welfare by changing their production or consumption.
- The market clearing price is the price of a market in equilibrium.

- Agent have access to public information.

In a competitive market, the equilibrium is also called the **competitive equilibrium** and the price is the **competitive price**.

Definition 3.5. The **Producer surplus** is the difference between the revenue of a producer and its variable cost. It is defined as:

$$PS_G = \lambda q_G(\lambda) - \int_0^{q_G(\lambda)} MC_G(x) dx \quad (3.6)$$

Definition 3.6. The **Consumer surplus** is the difference between the benefit of a consumer and the price they pay. It is defined as:

$$CS_L = \int_0^{q_L(\lambda)} MB_L(x) dx - \lambda q_L(\lambda) \quad (3.7)$$

Power markets

4.1 Exchange VS Pool market

Financial markets for electricity are a central part of the power system, they allow producers and consumers to buy and sell electricity and other services. There is 3 different ways to trade electricity:

- **Bilateral trade** are agreements between a producer and a consumer to buy/sell a certain amount of electricity at a certain price and time.
- **Exchange** is a centralized market where producers and consumers can buy and sell electricity by submitting simple bids with simple rules to auctions.
- **Pool** is a centralized market where producers and consumers can buy and sell electricity by submitting multi-part bids with complex rules to auctions.

Bilateral trade is the simplest way to trade electricity but it is the least centralised because all the are done in peer to peer. It is not efficient and thus cannot be used in real time. Exchange and pool are more centralised and thus more efficient.

4.1.1 Exchange Market

In an exchange market the bids look like this: (p_b, q_b) it is a pair of price and quantity. Positive values indicate a buy bid and negative values indicate a sell bid. The market operator collects all the bids then solves the following problem to determine the market clearing price and quantity:

$$\begin{aligned} \max_{\delta_b} \quad & \sum_{b \in B} p_b |q_b| \delta_b \\ \text{s.t.} \quad & \sum_{b \in B} q_b \delta_b = 0 \\ & 0 \leq \delta_b \leq 1 \quad \forall b \in B \end{aligned} \tag{4.1}$$

4.1.2 Pool market

In a pool market, the bids are more complex. For example, it can contains start-up costs, marginal costs and capacity. This would generate a triplet of information for each bid: (SC_g, MC_g, P_g) , in that case the market operator would solve the following

problem to determine the market clearing price and quantity:

$$\begin{aligned}
& \max_{d_l, p_g, u_g} \sum_{l \in L} MB_l d_l - \sum_{g \in G} (SC_g u_g + MC_g p_g) \\
& \text{s.t.} \quad \sum_{l \in L} d_l - \sum_{g \in G} p_g = 0 \\
& \quad 0 \leq d_l \leq D_l, \quad \forall l \in L \\
& \quad 0 \leq p_g \leq P_g u_g, \quad \forall g \in G \\
& \quad u_g \in \{0, 1\}, \quad \forall g \in G
\end{aligned} \tag{4.2}$$

4.2 Electricity auctions

In the electricity market, there exists different types of auctions.

Definition 4.1. The **uniform price auction** is an auction where an uniform price is decided λ , and every winning bid will pay or be paid at λ €/MWh. It is used in exchange markets.

Definition 4.2. The **second-price auction** is an auction where the price will be decided by the highest rejected bid.

Definition 4.3. The **first-price auction** is an auction where the price for each winning bid is the price of the bid itself.

Both the uniform price auction and the pay as bid auction are criticized for different reasons. **BUT IDK WHAT TO SAY HERE**

4.2.1 Blueprint of an electricity market

An electricity market is organized like this:

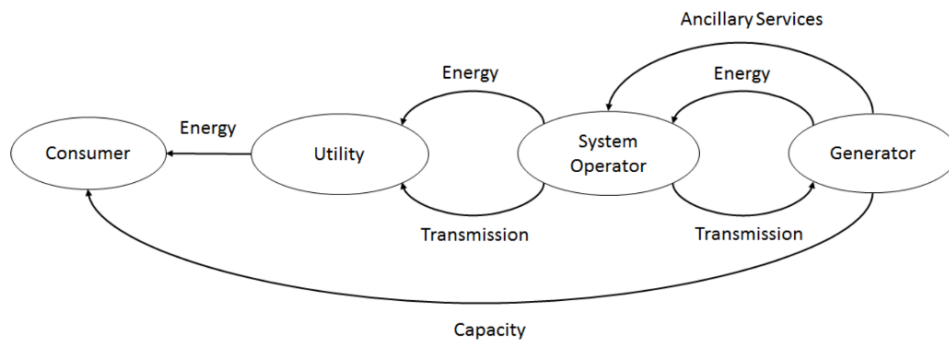


Figure 4.1: Blueprint of an electricity market

4.3 Nodal VS Zonal pricing

Until now, we have only considered market clearing problems without considering the physical configuration of the power system. In reality, the power system is a complex network of buses and lines, and the flow of electricity is determined by the laws of physics. This can lead to congestion in the lines. To manage this congestion and establish fair pricing, there are two main approaches: nodal pricing and zonal pricing.

4.3.1 Nodal pricing

Nodal pricing, is as the name suggests, a pricing mechanism where the price of electricity is determined at each node (or bus) in the power system. The price at each node reflects the marginal cost of supplying electricity to that node, this takes into account the cost of production, the cost of transmission and the congestion in the lines.

4.3.2 Zonal pricing

Zonal pricing, on the other hand, groups multiple nodes into zones and determines a single price for each zone. This approach is motivated by the need to simplify the pricing process and reduce the number of markets to be managed. This approach simplifies the pricing process but may not capture the full complexity of the power system and usually needs to be complemented with other mechanisms to manage congestion, such as financial transmission rights or redispatch.

Power flow

5.1 Network modeling

The power system can be modeled as a graph, where the nodes represent the buses and the edges represent the transmission lines. This graph can be characterized by some matrices. The one that interests us is the **Laplacian matrix** L , but to define it we need to define the **incidence matrix** A and the **degree matrix** D first. Let's consider a graph $G = (N, E)$, where N is the set of nodes and E is the set of edges.

Definition 5.1. The **incidence matrix** A is a matrix that describes the connectivity of the graph. It has one row for each node and one column for each node too. Each element of the matrix is defined as follows:

$$A_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases} \quad (5.1)$$

Definition 5.2. The **degree matrix** D is a diagonal matrix that contains the degree of each node. The degree of a node is the number of edges connected to it. It is defined as:

$$D_{ii} = \sum_j A_{ij} \quad (5.2)$$

Definition 5.3. The **Laplacian matrix** L is defined as the difference between the degree matrix and the incidence matrix:

$$L = D - A \quad (5.3)$$

It is a positive semi-definite matrix that captures the connectivity of the graph.

Theorem 5.4. The multiplicity of the eigenvalue $\lambda = 0$ in the Laplacian of a graph is equal to the number of connected components of the graph

5.2 Circuits

The state of an electrical circuit can be described by the voltage at each node and the current flowing through each edge. The edge have a passive role as electricity flows through them. And the nodes have an active role as they inject or consume electricity. To know the state of a circuit, we need to know the voltage at each node compared to a reference node (usually called the ground) and the current flowing through each edge.

5.3 The power flow equations

The power flow equations are based on the laws of physics, on the Kirchhoff's law.

Definition 5.5. The **Kirchhoff's current law** states that the total current entering a node must be equal to the total current leaving the node. Mathematically, it can be expressed as:

$$\sum I = 0 \quad (5.4)$$

Definition 5.6. The **Kirchhoff's voltage law** states that the total voltage around any closed loop in a circuit must be equal to zero. Mathematically, it can be expressed as:

$$\sum V = 0 \quad (5.5)$$

5.4 DC power flow using bus angles

TODO

5.5 DC power flow using lines flows

TODO

5.6 DC optimal power flow

Until now we have only considered optimization problems without considering the physical configuration of the power system, like in the economic dispatch problem. The DCOPF (DC Optimal Power Flow) problem, is an optimization problem that maximizes the total welfare like in the economic dispatch problem, but it also takes into account the physical configuration of the power system by including a linearization of the Kirchhoff's laws and the transmission limits constraints. To modelize that, there is two equivalent model. One based on the PTDF (Power Transfer Distribution Factor) and one based on the bus angles. Let us define how the transmission system is represented as a directed graph:

- N : set of nodes (buses)
- k : set of edges (lines) denoted $k = (m, n)$
- G_n : set of generators located at node n , $G = \cup_{n \in N} G_n$
- L_n : set of loads located at node n , $L = \cup_{n \in N} L_n$

The decision variables of the DCOPF problem will be p_g the amount of power produced by generator g , d_l the amount of power consumed by load l .

5.6.1 Power Transfer Distribution Factor model

For the PTDF model, we will interest us to the nodal injections in the model. For that, we will chose "hub node", which is a reference node that will absorb all the injections. The nodal injection at node n is defined as:

$$r_n = \sum_{g \in G_n} p_g - \sum_{l \in L_n} d_l \quad (5.6)$$

To follow the kirchhoff's laws, we need to add the following constraints:

$$\sum_{n \in N} r_n = 0 \quad (5.7)$$

Then, based on the PTDF (F_{kn}) that represents the amount of power flowing on line k as a result of shipping unit of power from node n to hub node, we can determine the flow on each line k as follows:

$$f_k = \sum_{n \in N} F_{kn} r_n \quad (5.8)$$

And we also need to add the transmission limits constraints:

$$-T_k \leq f_k \leq T_k \quad (5.9)$$

The full DCOPF model formulation with the PTDF is as follows:

$$\begin{aligned} & \max \sum_{l \in L} \int_0^{d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\ \text{s.t. } & f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^+] \\ & -f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^-] \\ & f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \quad [\psi_k] \\ & r_n - \sum_{g \in G_n} p_g + \sum_{l \in L_n} d_l = 0 \quad \forall n \in N \quad [\rho_n] \\ & \sum_{n \in N} r_n = 0 \quad [\phi] \\ & 0 \leq p_g \leq P_g \quad \forall g \in G \quad [\mu_g] \\ & 0 \leq d_l \leq D_l \quad \forall l \in L \quad [\nu_l] \end{aligned} \quad (5.10)$$

TO CHECK The KKT conditions for this problem are:

$$\begin{aligned} & 0 \leq p_g \perp MC_g(p_g) - \rho_{n(g)} + \mu_g \geq 0 \\ & 0 \leq d_l \perp -MB_l(d_l) + \rho_{n(l)} + \nu_l \geq 0 \\ & 0 \leq f_k \perp T_k - f_k \geq 0 \\ & 0 \leq -f_k \perp T_k + f_k \geq 0 \\ & 0 \leq \mu_g \perp P_g - p_g \geq 0 \\ & 0 \leq \nu_l \perp D_l - d_l \geq 0 \\ & f_k \perp f_k - \sum_{n \in N} F_{kn} r_n = 0 \\ & r_n \perp r_n - \sum_{g \in G_n} p_g + \sum_{l \in L_n} d_l = 0 \end{aligned} \quad (5.11)$$

With these, we can see that there exists a threshold ρ_n for all n such that:

- If $0 < p_g < P_g$, then $\rho_n = MC_g(p_g)$
- If $0 < d_l < D_l$, then $\rho_n = MB_l(d_l)$
- If $p_g = 0$, then $\rho_n \leq MC_g(p_g)$
- If $d_l = 0$, then $\rho_n \geq MB_l(d_l)$
- If $p_g = P_g$, then $\rho_n \geq MC_g(p_g)$
- If $d_l = D_l$, then $\rho_n \leq MB_l(d_l)$

TO CHECK We can interpret the dual variables as follows:

- ϕ induce marginal changes in welfare from marginal increase in production or marginal decrease in consumption.
- λ_k^+ and λ_k^- induces marginal impact of increasing line capacity.
- ρ_n induces marginal impact of marginal increase in consumption or marginal decrease in production at node n .

5.6.2 Bus angle model

For the bus angle model, also known as the reactance model, we will interest us to the differences of phase angles in the network. For that, we will chose a "hub node", its choice is independent. The phase angle at node n is denoted θ_n . The flow on line $k = (m, n)$ is defined as:

$$f_{(m,n)} = B_{(m,n)}(\theta_m - \theta_n) \quad (5.12)$$

Where $B_{(m,n)}$ is the susceptance of line (m, n) and $\theta_{hub} = 0$. Then, we need to define the conservation of energy constraints at each node:

$$\sum_{g \in G_n} p_g - \sum_{k=(.,n)} f_k = \sum_{k=(n,.)} f_k + \sum_{l \in L_n} d_l \quad (5.13)$$

The full DCOPF model formulation with the bus angle is as follows:

$$\begin{aligned} & \max \sum_{l \in L} \int_0^{d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\ \text{s.t. } & f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^+] \\ & -f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^-] \\ & f_{(m,n)} - B_{(m,n)}(\theta_m - \theta_n) = 0 \quad \forall (m,n) \in K \quad [\gamma_{(m,n)}] \\ & \sum_{g \in G_n} p_g - \sum_{k=(.,n)} f_k + \sum_{k=(n,.)} f_k - \sum_{l \in L_n} d_l = 0 \quad [\rho_n] \\ & 0 \leq p_g \leq P_g \quad \forall g \in G \quad [\mu_g] \\ & 0 \leq d_l \leq D_l \quad \forall l \in L \quad [\nu_l] \end{aligned} \quad (5.14)$$

Pricing transmission

In the DCOPF problem, we have seen that there exists a different price ρ_n for each node n , this different price at each node is induced by the transmission constraints. Intuitively, if a node has cheap generation and cannot export much, its price will be low and if a node has high demand cannot import enough cheap electricity, its price will be high. With that we can introduce a new market operator.

Definition 6.1. The **Transmission market operator (TSO)** is the market operator that manages the grid. It can be seen as an agent that buys electricity where it is cheap and sells it where it is expensive while respecting the transmission constraints.

With that, we can rewrite the DCOPF problem as a decentralized problem with different market operators: producers, consumers, TSO and the auctioneer. Each of these agents will solve their own optimization problem and the market clearing condition will be the same as before. The TSO will solve the following problem:

$$\begin{aligned}
 \max \quad & \sum_{n \in N} -\rho_n r_n \\
 \text{s.t.} \quad & f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \\
 & \sum_{n \in N} r_n = 0 \\
 & -T_k \leq f_k \leq T_k \quad \forall k \in K
 \end{aligned} \tag{6.1}$$

The producers will solve the following problem:

$$\begin{aligned}
 \max \quad & p_g \rho_{n(g)} - \int_0^{p_g} MC_g(x) dx \\
 \text{s.t.} \quad & 0 \leq p_g \leq P_g \quad \forall g \in G
 \end{aligned} \tag{6.2}$$

The consumers will solve the following problem:

$$\begin{aligned}
 \max \quad & \int_0^{d_l} MB_l(x) dx - d_l \rho_{n(l)} \\
 \text{s.t.} \quad & 0 \leq d_l \leq D_l \quad \forall l \in L
 \end{aligned} \tag{6.3}$$

And the auctioneer will solve the following problem:

$$\max \quad \rho_n (r_n + \sum_{l \in L_n} d_l - \sum_{g \in G_n} p_g) \tag{6.4}$$

But solving all these problems requires a lot of communication between the different agents, and thus it is not very practical. Plus the TSO needs to take position in the different markets and has lots of opportunities to abuse of its position.

6.1 Nodal pricing

In nodal pricing also called local marginal pricing (LMP), there is an uniform price auction then the TSO will solve the DCOPF that will decide the price ρ_n at each node n . If all the agents bid truthfully, then the solution outputed by the LMP, will be the same as the optimal solution of the DCOPF problem. In this configuration, if there is no congestion, prices will tend to be equals at all nodes, but if there is congestion, prices will differ between nodes.

The congestion on the lines and the choices made by the TSO to manage this congestion generates a cost for the consumers and a rent for the producers.

Definition 6.2. The **congestion rent** is the profit that the TSO makes by buying electricity where it is cheap and selling it where it is expensive. It is defined as:

$$\sum_{n \in N} \rho_n \left(\sum_{l \in L_n} d_l - \sum_{g \in G_n} p_g \right) \quad (6.5)$$

6.2 Zonal pricing

Nodal pricing is criticized for being too complex because it generates too many prices. Plus the multiplicity of small markets leads to a lack of liquidity in each market but also to a lot of opportunities for market power abuse. So an alternative to nodal pricing is zonal pricing, where the nodes are grouped into zones and a single price is determined for each zone. The zonal DCOPF problem is formulated as follows:

$$\begin{aligned} \max \quad & \sum_{l \in L} \int_0^{d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\ \text{s.t.} \quad & f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^+] \\ & -f_k \leq T_k \quad \forall k \in K \quad [\lambda_k^-] \\ & f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \quad [\psi_k] \\ & r_z - \sum_{g \in G_z} p_g + \sum_{l \in L_z} d_l = 0 \quad \forall z \in Z \quad [\rho_z] \\ & \sum_{n \in N} r_n = 0 \quad [\phi] \\ & 0 \leq p_g \leq P_g \quad \forall g \in G \quad [\mu_g] \\ & 0 \leq d_l \leq D_l \quad \forall l \in L \quad [\nu_l] \end{aligned} \quad (6.6)$$

In zonal pricing, there is an uniform price auction then the TSO will solve the zonal DCOPF that will decide the price ρ_z at each zone z . Within each zone, Kirchhoff's laws are ignored this can cause congestion problems. These congestion problems needs to be solved by the TSO before the time of delivery by redispatching the production and consumption. The redispatch works with a pay as bid auction, where the market participants are forced to bid their true marginal cost/benefit. The redispatch is thus financially neutral to market participants. The TSO will cover the difference between payments received and made. The **perfect cost-based redispatch** minimize the

redispatch cost for the TSO and given the dispatch obtained under the zonal pricing (p_g^*, d_l^*) , it is formulated as follows:

$$\begin{aligned}
RD = \max & \sum_{l \in L} \int_0^{d_l^* + \Delta d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g^* + \Delta p_g} MC_g(x) dx \\
\text{s.t. } & f_k \leq T_k \quad \forall k \in K \\
& -T_k \leq f_k \leq T_k \quad \forall k \in K \\
& P_g - p_g^* - \Delta p_g \quad \forall g \in G \\
& D_l - d_l^* - \Delta d_l \quad \forall l \in L \\
& r_n + \sum_{g \in G_n(n)} (p_g^* + \Delta p_g) - \sum_{l \in L_n(n)} (d_l^* + \Delta d_l) = 0 \quad \forall n \in N \\
& f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \\
& \sum_{n \in N} r_n = 0
\end{aligned} \tag{6.7}$$

6.3 Other formulations

6.3.1 Zonal pricing with Availability Transfer Capacities (ATC)

The formulation of the zonal pricing with ATC is the same as the one of the zonal pricing, but with an additional constraint that limits the flow between zones. The full formulation is as follows:

$$\begin{aligned}
\max & \sum_{l \in L} \int_0^{d_l} MB_l(x) dx - \sum_{g \in G} \int_0^{p_g} MC_g(x) dx \\
\text{s.t. } & \sum_{a=(z, \cdot)} f_a + \sum_{l \in L_z} d_l = \sum_{a=(\cdot, z)} f_a + \sum_{g \in G_z} p_g \quad \forall z \in Z \quad [\rho_z] \\
& -ATC_a \leq f_a \leq ATC_a \quad \forall a \in A \quad [\lambda_a^+, \lambda_a^-] \\
& 0 \leq p_g \leq P_g \quad \forall g \in G \quad [\mu_g] \\
& 0 \leq d_l \leq D_l \quad \forall l \in L \quad [\nu_l]
\end{aligned} \tag{6.8}$$

Where A is the set of inter-zonal lines and ATC_a is the capacity of line a that between two zones.

6.3.2 Flow-based (FB) zonal pricing

Unlike the zonal pricing with ATC, the flow-based zonal use the PTDF to model the flow between zones and thus the flow between zones influences the flow between other zones. Using zonal PTDF, the constraints of the flow between zones are as follows:

$$-T_k \leq \sum_{z \in Z} \tilde{F}_{kz} \left(\sum_{g \in G_z} p_g - \sum_{l \in L_z} d_l \right) \leq T_k \quad \forall k \in K \tag{6.9}$$

We need to define the zonal PTDF $\tilde{F}$, it is defined as follows:

$$\tilde{F}_{kz} = \sum_{n \in N} \frac{\tilde{r}_n}{\sum_{n \in N} \tilde{r}_n} F_{kn} \tag{6.10}$$

So the contribution of a node to the zonal PTDF is proportional to its nodal injection to the total nodal injection of the zone.

Unit commitment and nonconvexities

Until now, we considered only basic convex problems, like in the economic dispatch. But in reality, real power plant are much more complex systems and their cost functions introduces non-convexities. These non-convexities are for examples start-up costs, no load costs, minimum up and down times, ramping constraints, minimum stable generation, etc. To address these non-convexities, we introduce the unit commitment problem.

7.1 The unit commitment problem

The unit commitment problem is an optimization problem that determines the optimal schedule of power plants to meet the demand while minimizing the total cost. For that, we need to introduce times dependant variables:

- $p_{g,t}$: power output of generator g at time t
- $u_{g,t}$: binary variable that indicates whether generator g is on or off at time t
- $v_{g,t}$: binary variable that indicates whether generator g is starting up at time t
- $w_{g,t}$: binary variable that indicates whether generator g is shutting down at time t

This model with non-convexities uses binary variables and thus is a mixed-integer linear program (MILP). The formulation of the unit commitment problem is as follows:

$$\begin{aligned}
 \min \sum_g \sum_t (MC_g p_{g,t} + K_g u_{g,t} + S_g v_{g,t}) \\
 \text{s.t.} \quad & \sum_g p_{g,t} = D_t \quad \forall t \\
 & P_g^{\min} u_{g,t} \leq p_{g,t} \leq P_g^{\max} u_{g,t} \quad \forall g, t \\
 & u_{g,t} = u_{g,t-1} + v_{g,t} - w_{g,t} \quad \forall g, t \\
 & u_{g,t}, v_{g,t}, w_{g,t} \in \{0, 1\} \quad \forall g, t
 \end{aligned} \tag{7.1}$$

For additional ramp constraint, we can add the following constraints:

$$\begin{aligned}
 p_{g,t} &\leq p_{g,t-1} + R_g^+ - v_{g,t}(R_g^+ - R_{SU}^+) \quad \forall t \\
 p_{g,t} &\geq p_{g,t-1} - R_g^- + w_{g,t}(R_g^- - R_{SD}^-) \quad \forall t
 \end{aligned} \tag{7.2}$$

where R_g^+ and R_g^- are the ramp up and down limits of generator g , R_{SU}^+ and R_{SD}^- are the ramp up and down limits during start-up and shut-down respectively. For additional minimum up and down time constraints, we can add the following constraints:

$$\begin{aligned} \sum_{i=t-UT_g+1}^t v_{g,i} &\leq u_{g,t} \quad \forall t \geq UT_g \\ \sum_{i=t-DT_g+1}^t w_{g,i} &\leq 1 - u_{g,t} \quad \forall t \geq DT_g \end{aligned} \quad (7.3)$$

Due to the heaviness of the unit commitment problem, and its time dependancies, the model needs to be solved ahead of time and thus it is used for the day-ahead market. Usually, it is solved once for the whole next day.

7.2 Market equilibrium under non-convexities

Finding the market equilibrium under non-convexities is a complex problem. Ideally, we want to obtain the four fundamental properties of markets.

- **Efficiency:** the market equilibrium should maximize the total welfare of the market.
- **Individual rationality:** the agents have no incentive to deviate from the market equilibrium.
- **Cost recovery:** every agent has a non-negative profit
- **Revenue adequacy:** the revenue of the market operator is non-negative

The **Impossibility theorem** states that:

Theorem 7.1. Once you introduce the discrete nature of power production, it becomes mathematically impossible to find fundamental prices that respect the 4 properties.

There exists multiples relaxation of the unit commitment problem that can be used to solve a convexified version of the problem.

7.2.1 Marginal pricing

The marginal pricing is the simplest way to convexify the problem, it consists in solving the unit commitment problem and then fixing the unit commitment variables to their optimal value and then solving the resulting convex problem to find the marginal prices. But the solution will not be an equilibrium. Some agent could still have financial incentives to deviate from the solution. We can measure these financial incentives by the **lost opportunity cost** of an agent.

Definition 7.2. The **lost opportunity cost** of an agent is the difference between the maximum possible profit of the agent and it's profit as the market clears.

$$LOC_g(\pi_t) = \max \sum_t (\pi_t - MC_g) p_{g,t} - K_g u_{g,t} - S_g v_{g,t} - \sum_t (\pi_t - MC_g) p_{g,t}^* - K_g u_{g,t}^* - S_g v_{g,t}^* \quad (7.4)$$

with $(p_{g,t}^*, u_{g,t}^*, v_{g,t}^*)$ the optimal solution of the unit commitment problem and π_t the decided price at time t .

7.2.2 Convex hull pricing

Convex hull pricing is a more complex way to find a market equilibrium under non-convexities. It consists in computing the optimal lagrange multipliers of the market clearing constraints in the nonconvex problem.

Theorem 7.3. The Hogan theorem states that the convex hull prices are the uniform prices that minimize the total lost opportunity costs.

The solution of the convex hull pricing will lead to a dispatch that is the same as the one of the unit commitment problem, but with different prices. Like the marginal pricing, the dispatch obtained are welfare maximizing.

7.2.3 Euphemia pricing

TODO

7.2.4 Comparison between models

In the start of the section, we have seen that it is impossible to find a market equilibrium that respect the 4 fundamental properties of markets. But they can respect some of them. Let's compare the different models:

Model	MP	MP with uplifts	CHP with uplifts	EUPHEMIA pricing
Efficiency	Yes	Yes	Yes	No
Individual Rationality	No	Yes	Yes	No
Cost Recovery	No	Yes	Yes	Yes
Revenue Adequacy	Yes	No	No	Yes

Cost recovery is very important for incentivizing market participation, which is why uplifts are used.